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B.Tech. Degree I Semester Examination November 2019

CS/EC/IT 19-200-0105 B BASIC MECHANICAL ENGINEERING
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) Write a note on thermodynamic property. Why work is not considered as a property?
- (b) "Carnot cycle is impractical and calculation of Carnot efficiency is insignificant". Analyze the statement and comment on it.
- (c) With a neat sketch explain the working of a carburetor.
- (d) Differentiate between heat pump and a refrigerator. Establish the connection between the CoP of refrigerator and heat pump.
- (e) Define specific speed of a turbine. What is its significance?
- (f) Explain the difference between positive displacement and rotodynamic pump. Give examples.
- (g) What is heat treatment? What are the purposes of heat treatment?
- (h) What are the classifications of gears? Explain.

PART B

(4 × 12 = 48)

- II. What do you mean by a thermodynamic process? Find the work done and internal energy change for an: (i) isochoric process (ii) isothermal process (iii) adiabatic process. (12)
- OR**
- III. (a) Explain first law of thermodynamics. Apply first law for an open system. (6)
- (b) Find the difference between work done in compressing 0.28 m³ of air at a pressure of 1.4 bar to a volume of 0.028 m³, when the compression is adiabatic and polytropic with polytropic index $n=1.2$. (6)
- IV. Explain the working of 2 stroke SI engine. What are the advantages and disadvantages of a 2-stroke engine over a 4 stroke engine? (12)
- OR**
- V. Explain diesel cycle. Obtain the expression for air standard efficiency of a diesel cycle. (12)
- VI. (a) Distinguish between impulse and reaction turbine. (4)
- (b) With the help of a neat sketch explain the working of Pelton turbine. (8)
- OR**
- VII. (a) Describe the working of a thermal power plant with suitable layout diagram. (8)
- (b) What are the advantages and disadvantages of following power plants. (4)
 - (i) Hydroelectric
 - (ii) Nuclear
- VIII. (a) Explain the process of gas welding using suitable diagrams. (6)
- (b) What is meant by sand casting? Explain sand casting procedure. (6)
- OR**
- IX. (a) Derive an expression for length of belt for open belt drive. (8)
- (b) Explain with diagrams (4)
 - (i) Extrusion
 - (ii) Rolling
